SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Scenario-Based Questions

Course Code:	**DSA0405**	**Course Title:**	**FUNDAMENTALS OF DATA SCIENCE**
CO Assessed:	**CO4 – Scenario-Based Questions**	**Assessment:**	**Assessment Tool 1 – Scenario-Based Questions**
Weightage:	**40% of CIA**	**Total Marks:**	**100**
CO4 Statement:	To apply the statistical inferences such as testing of hypothesis and point estimates for the given data		
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Section / Batch:	2024	**Date:**	13/8/2026

Code of Conduct

I Kishore Kumar. P(192424108) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Kshore Kumar. P

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**	**Marks**
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

CO4-AT1 – Scenario-Based Questions

Detailed Answers

1. Food Delivery Company – Point Estimate

Question:

A food delivery company wants to know the average delivery time in Chennai. It randomly selects 100 deliveries and obtains an average delivery time of 32 minutes. What population parameter is being estimated, and what is the point estimate?

Answer:

The company wants to estimate the average delivery time of all food deliveries in Chennai.

Let:

- Population: All food deliveries made in Chennai
- Population parameter: Population mean, represented by μ
- Sample: 100 randomly selected deliveries
- Sample mean: 32 minutes

The population mean μ is unknown because it is not practical to collect the delivery time of every delivery in Chennai. Therefore, the sample is used to estimate it.

The sample mean is calculated as:

$$\bar{x} = \Sigma x / n$$

Here, the given sample mean is already:

$$\bar{x} = 32 \text{ minutes}$$

In point estimation, the sample mean is used as the best single estimate of the population mean.

Population parameter being estimated:

$$\mu$$

where μ represents the average delivery time of all deliveries in Chennai.

Point estimate:

$$\bar{x} = 32 \text{ minutes}$$

Interpretation: The best single estimate of the average delivery time for all deliveries in Chennai is 32 minutes.

2. E-Commerce Website – Point Estimate of Population Proportion

Question:

An e-commerce website wants to estimate the percentage of customers who purchase a product after viewing its advertisement. From 800 randomly selected visitors, 176 made a purchase. Determine the point estimate of the population proportion.

Answer:

The company wants to estimate the proportion of all visitors who purchase the product after viewing the advertisement.

Let:

- Population: All visitors who view the advertisement
- Sample size: $n = 800$
- Number who made a purchase: $x = 176$
- Population proportion: p
- Sample proportion: $\hat{p}$

The point estimate of the population proportion is the sample proportion.

The formula is:

$$\hat{p} = x / n$$

Substituting the given values:

$$\hat{p} = 176 / 800$$

$$\hat{p} = 0.22$$

To express this as a percentage:

$$0.22 \times 100 = 22\%$$

Therefore, the point estimate is:

$$\hat{p} = 0.22 \text{ or } 22\%$$

Interpretation: Based on the sample, approximately 22% of all visitors are estimated to purchase the product after viewing the advertisement.

3. Bank – Frequentist Approach to Estimating the Population Mean

Question:

A bank wants to estimate the average monthly expenditure of its credit-card customers. A random sample of 150 customers has a mean expenditure of ₹24,500. Explain how the sample mean can be used as a frequentist estimate of the population mean.

Answer:

The bank wants to estimate the average monthly expenditure of all its credit-card customers.

Let:

- Population: All credit-card customers of the bank
- Population mean: μ
- Sample size: $n = 150$
- Sample mean: $\bar{x} = ₹24,500$

In the frequentist approach, the population parameter is treated as a fixed but unknown quantity. The sample is considered random, and statistics calculated from the sample are used to estimate the unknown population parameter.

The sample mean is calculated using:

$$\bar{x} = \Sigma x_i / n$$

The problem gives:

$$\bar{x} = ₹24{,}500$$

The sample mean is used as the point estimate of the population mean.

$$\mu \approx \bar{x} = ₹24{,}500$$

The reasoning is that, when the sample is randomly selected and sufficiently representative of the population, the sample mean provides an unbiased estimate of the population mean.

In the frequentist view, if many random samples of 150 customers were repeatedly selected, each sample would produce a different sample mean. These sample means would form a sampling distribution centered around the true population mean.

Thus, ₹24,500 is our estimate of the unknown population average.

Conclusion: Frequentist estimate of population mean = ₹24,500.

Interpretation: The bank estimates that the average monthly expenditure of all its credit-card customers is approximately ₹24,500.

4. Streaming Platform – Point Estimate of Population Proportion

Question:

A streaming platform selects 500 subscribers to estimate the percentage of users who watch movies every day. If 320 users report watching movies daily, calculate the point estimate and explain its interpretation.

Answer:

The company wants to estimate the proportion of all subscribers who watch movies every day.

Let:

- Population: All subscribers of the streaming platform
- Sample size: $n = 500$
- Number of daily movie watchers: $x = 320$
- Population proportion: p
- Sample proportion: $\hat{p}$

The point estimate of a population proportion is the sample proportion.

The formula is:

$$\hat{p} = x / n$$

Substitute the values:

$$\hat{p} = 320 / 500$$

$$\hat{p} = 0.64$$

Convert it into percentage:

$$0.64 \times 100 = 64\%$$

Therefore:

$$\hat{p} = 0.64 \text{ or } 64\%$$

Interpretation: From the sample, the platform estimates that approximately 64% of all subscribers watch movies every day.

Since the 500 subscribers were selected as a sample, 64% is the best single estimate of the corresponding population proportion.

5. Smartphone Battery Life – 95% Confidence Interval

Question:

A manufacturing company randomly selects 64 smartphones and finds that their average battery life is 18.5 hours with a standard deviation of 2 hours. Construct a 95% confidence interval for the average battery life.

Answer:

We need to estimate the population mean battery life using a 95% confidence interval.

Given data:

- $n = 64$
- $\bar{x} = 18.5$ hours
- $s = 2$ hours
- Confidence level = 95%

Since the population standard deviation is not given and the sample standard deviation is provided, we use the t-distribution.

Degrees of freedom:

$$df = n - 1 = 64 - 1 = 63$$

For a 95% confidence interval with 63 degrees of freedom:

$$t_{0.025,63} \approx 1.998$$

Step 1: Find the standard error

The standard error of the sample mean is:

$$SE = s / \sqrt{n}$$

Substitute the values:

$$SE = 2 / \sqrt{64} = 2 / 8 = 0.25 \text{ hours}$$

Therefore, SE = 0.25 hours.

Step 2: Find the margin of error

The formula is:

$$\mathbf{ME = t\alpha/2 \times SE}$$

Substituting:

$$\mathbf{ME = 1.998 \times 0.25 = 0.4995\ hours}$$

Approximately, ME ≈ 0.50 hours.

Step 3: Construct the confidence interval

The confidence interval formula is:

$$\mathbf{\bar{x} \pm ME}$$

Therefore:

$$\mathbf{18.5 \pm 0.4995}$$

Lower limit:

$$\mathbf{18.5 - 0.4995 = 18.0005\ hours}$$

Upper limit:

$$\mathbf{18.5 + 0.4995 = 18.9995\ hours}$$

Hence, the 95% confidence interval is:

$$\mathbf{(18.0005, 18.9995)\ hours}$$

Approximately:

$$\mathbf{(18.00, 19.00)\ hours}$$

Interpretation: We are 95% confident that the true average battery life of the population of smartphones lies between approximately 18.00 hours and 19.00 hours.

Final answer:

$$\mathbf{18.00\ hours < \mu < 19.00\ hours}$$

where μ is the true population mean battery life.

Final Answers at a Glance

Question	Concept	Final Answer
1	Point Estimate	Population parameter = μ; Point estimate = 32 minutes
2	Point Estimate	$\hat{p}$ = 176/800 = 0.22 = 22%
3	Frequentist Approach	$\bar{x}$ = ₹24,500 is the estimate of μ
4	Point Estimate	$\hat{p}$ = 320/500 = 0.64 = 64%
5	Confidence Interval	95% CI ≈ (18.00, 19.00) hours